Homework1

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1 AI Use and Verification Statement

I used ChatGPT (OpenAI, GPT-6; accessed September 29, 2026) to help draft this AI-use statement. I checked the statement against the homework instructions. ChatGPT has not been used to generate the answers or R calculations reported in this submission.

2 Question 1

2.1 Question 1.1

Let $G_1 = \{j : X_j = 0\}$ and $G_2 = \{j : X_j = 1\}$, where $\sum_{j \in G_1} = n_1$ and $\sum_{j \in G_2} =$ n_2$. The model gives

$$\mu_1 = \beta_0, \qquad \mu_2 = \beta_0 + \beta^*.$$

$$H_0 : \mu_1 = \mu_2 \equiv H_0 : \beta^* = 0.$$

$$L(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2) = \prod_{j \in G_1} \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{ -\frac{(Y_j - \beta_0)^2}{2\sigma_1^2} \right\} \prod_{j \in G_2} \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{ -\frac{(Y_j - \beta_0 - \beta^*)^2}{2\sigma_2^2} \right\}.$$

$$\ell(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2) = -\frac{n_1 + n_2}{2} \log(2\pi) - \frac{n_1}{2} \log(\sigma_1^2) - \frac{n_2}{2} \log(\sigma_2^2) - \frac{1}{2\sigma_1^2} \sum_{j \in G_1} (Y_j - \beta_0)^2 - \frac{1}{2\sigma_2^2} \sum_{j \in G_2} (Y_j - \beta_0 - \beta^*)^2.$$

Treating σ_1^2 and σ_2^2 as known, the score equations are

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma_1^2} \sum_{j \in G_1} (Y_j - \beta_0) + \frac{1}{\sigma_2^2} \sum_{j \in G_2} (Y_j - \beta_0 - \beta^*),$$
$$\frac{\partial \ell}{\partial \beta^*} = \frac{1}{\sigma_2^2} \sum_{j \in G_2} (Y_j - \beta_0 - \beta^*).$$

Setting the scores to zero gives

$$\hat{\beta}_0 = \overline{Y}_1, \qquad \hat{\beta}^* = \overline{Y}_2 - \overline{Y}_1.$$

The negative Hessian, which is also the Fisher information for $(\beta_0, \beta^*)^T$, is

$$I = \begin{pmatrix} \frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2} & \frac{n_2}{\sigma_2^2} \\ \frac{n_2}{\sigma_2^2} & \frac{n_2}{\sigma_2^2} \end{pmatrix}.$$

With the equation of invertion

$$V = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} = \frac{1}{ac - b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}, \qquad ac - b^2 \neq 0.$$

$$I^{-1} = \begin{pmatrix} \frac{\sigma_1^2}{n_1} & -\frac{\sigma_1^2}{n_1} \\ -\frac{\sigma_1^2}{n_1} & \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \end{pmatrix}.$$

Thus Wald statistic is

$$Z_W = \frac{\hat{\beta}^*}{\sqrt{\mathrm{Var}(\hat{\beta}^*)}} = \frac{\overline{Y}_2 - \overline{Y}_1}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}.$$

$$W = Z_W^2 = \frac{(\overline{Y}_2 - \overline{Y}_1)^2}{\sigma_1^2/n_1 + \sigma_2^2/n_2}.$$

2.2 Question 1.2

$$S_g = \sum_{j \in G_g} (Y_j - \overline{Y}_g)^2, g = 1, 2.$$

$$\hat{\beta}_0 = \overline{Y}_1, \ \hat{\beta}^* = \overline{Y}_2 - \overline{Y}_1, \ \hat{\sigma}_{g,\mathrm{ML}}^2 = \frac{S_g}{n_g}.$$

$$\begin{aligned} W_{\mathrm{ML}} &= \frac{(\hat{\beta}^*)^2}{\hat{\sigma}_{1,\mathrm{ML}}^2/n_1 + \hat{\sigma}_{2,\mathrm{ML}}^2/n_2} \\ &= \frac{(\overline{Y}_2 - \overline{Y}_1)^2}{S_1/n_1^2 + S_2/n_2^2}. \end{aligned}$$

plug in:

$$s_g^2 = \frac{S_g}{n_g - 1}.$$

$$Z^2 = \frac{(\overline{Y}_2 - \overline{Y}_1)^2}{s_1^2/n_1 + s_2^2/n_2}$$
$$= \frac{(\overline{Y}_2 - \overline{Y}_1)^2}{S_1/[n_1(n_1-1)] + S_2/[n_2(n_2-1)]}.$$

$$W_{\mathrm{ML}} \neq Z^2$$

For any finite n_1,n_2.

$$W_{\mathrm{Wald}}(s_1^2, s_2^2) = \frac{(\hat{\beta}^*)^2}{s_1^2/n_1 + s_2^2/n_2} = Z^2.$$

The difference vanishes asymptotically as both group sizes grow. Equal group sizes alone do not make the finite-sample statistics equal.

$$\frac{\hat{\sigma}_{g,\mathrm{ML}}^2}{s_g^2} = \frac{n_g - 1}{n_g} \longrightarrow 1 \quad n_g \longrightarrow \infty.$$

3 Question 2

For $X_j = 0$, let $p_1 = \beta_0$.

For $X_j = 1$, let $p_2 = \beta_0 + \beta^*$.

$$H_0 : p_1 = p_2 \;\equiv\; H_0 : \beta^* = 0.$$

Let $G_1 = \{j : X_j = 0\}$ and $G_2 = \{j : X_j = 1\}$. The likelihood is

$$L(\beta_0, \beta^*) = \prod_{j \in G_1} \beta_0^{Y_j}(1-\beta_0)^{1-Y_j} \prod_{j \in G_2} (\beta_0+\beta^*)^{Y_j}(1-\beta_0-\beta^*)^{1-Y_j}.$$

The log-likelihood is

$$\ell(\beta_0, \beta^*) = \sum_{j \in G_1} \left[Y_j \log(\beta_0) + (1-Y_j)\log(1-\beta_0)\right] + \sum_{j \in G_2} \left[Y_j \log(\beta_0+\beta^*) + (1-Y_j)\log(1-\beta_0-\beta^*)\right]$$

$$\begin{aligned}
\frac{\partial \ell}{\partial \beta_0} &= \sum_{j \in G_1} \frac{Y_j - p_1}{p_1(1-p_1)} + \sum_{j \in G_2} \frac{Y_j - p_2}{p_2(1-p_2)}, \\
\frac{\partial \ell}{\partial \beta^*} &= \sum_{j \in G_2} \frac{Y_j - p_2}{p_2(1-p_2)}.
\end{aligned}$$

Setting both scores to zero yields

$$\begin{aligned}
\hat{p}_1 &= \frac{1}{n_1} \sum_{j=1}^{n_1+n_2} (1-X_j)Y_j, \\
\hat{p}_2 &= \frac{1}{n_2} \sum_{j=1}^{n_1+n_2} X_j Y_j, \\
\hat{\beta}_0 &= \hat{p}_1, \\
\hat{\beta}^* &= \hat{p}_2 - \hat{p}_1.
\end{aligned}$$

The Fisher information matrix is

$$I(\beta_0, \beta^*) = \begin{pmatrix} \dfrac{n_1}{p_1(1-p_1)} + \dfrac{n_2}{p_2(1-p_2)} & \dfrac{n_2}{p_2(1-p_2)} \\ \dfrac{n_2}{p_2(1-p_2)} & \dfrac{n_2}{p_2(1-p_2)} \end{pmatrix}.$$

With the equation of invertion

$$V = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} = \frac{1}{ac - b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}, \qquad ac - b^2 \neq 0.$$

$$I(\beta_0, \beta^*)^{-1} = \frac{p_1(1-p_1)p_2(1-p_2)}{n_1 n_2} \begin{pmatrix} \dfrac{n_2}{p_2(1-p_2)} & -\dfrac{n_2}{p_2(1-p_2)} \\ -\dfrac{n_2}{p_2(1-p_2)} & \dfrac{n_1}{p_1(1-p_1)} + \dfrac{n_2}{p_2(1-p_2)} \end{pmatrix}$$

$$= \begin{pmatrix} \dfrac{p_1(1-p_1)}{n_1} & -\dfrac{p_1(1-p_1)}{n_1} \\ -\dfrac{p_1(1-p_1)}{n_1} & \dfrac{p_1(1-p_1)}{n_1} + \dfrac{p_2(1-p_2)}{n_2} \end{pmatrix}.$$

Hence, the variance function is:

$$\widehat{\mathrm{Var}}(\hat{\beta}^*) = \frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}.$$

The Wald statistic for (H_0: ^*=0) is

$$\begin{aligned} W &= \frac{(\hat{\beta}^*)^2}{\widehat{\mathrm{Var}}(\hat{\beta}^*)} \\ &= \frac{(\hat{p}_2 - \hat{p}_1)^2}{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2} \\ &\equiv Z^2 \end{aligned}$$

This is exactly the $W \equiv Z^2$ statistic specified in the question. Under H_0 and the usual large-sample regularity conditions, W has an asymptotic χ^2_1 distribution, so as Z^2.

4 Question 3

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i, \qquad H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0.$$

```r
load(here::here("bis629_data", "MaxArt_HW1_2026.RData"))
data0 <- MaxArt_HW1_2026

# summary(data0)
# View(data0)

## H_0: beta1 = 0
## versus
## HA: beta1 != 0
mod0 <- glm(Y ~ 1, data = data0, family = binomial())
mod1 <- glm(Y ~ 1 + X, data = data0, family = binomial())

## 1. Wald test
beta <- unname(coef(mod1)["X"])
se <- unname(coef(summary(mod1))["X", "Std. Error"])
wald <- (beta / se)^2

## 2. Score test
score_output <- anova(mod0, mod1, test = "Rao")
score <- score_output$Rao[2]

## 3. Likelihood-ratio test
lrt <- 2 * as.numeric(logLik(mod1) - logLik(mod0))

## Comparison table
stats <- c(wald, score, lrt)

results <- data.frame(Test = c("Wald", "Score", "LRT"),
                      Chi_square = stats,
                      df = 1,
                      p_value = pchisq(stats, df = 1, lower.tail = FALSE))
```

```r
print(results, digits = 4, row.names = FALSE)
```

```
#>   Test Chi_square df p_value
#>   Wald      1.072  1  0.3005
#>  Score      1.079  1  0.2989
#>    LRT      1.070  1  0.3008
```

In general the LRT test is the most powerful one among the three. It directly compares the maximized likelihoods of the models and is often preferable to the Wald test.

5 Question 4

```r
# Parameters estimated from MaxART
n <- nrow(data0)
p_x <- mean(data0$X)
b0 <- unname(coef(mod1)["(Intercept)"])
b1 <- unname(coef(mod1)["X"])
alpha <- 0.05
critical <- qchisq(1 - alpha, df = 1)
tests <- c("Wald", "Score", "LRT")

ncp_all <- function(n, b1) {
  p0 <- plogis(b0)
  p1 <- plogis(b0 + b1)
  v0 <- p0 * (1 - p0)
  v1 <- p1 * (1 - p1)

  # Wald
  Ib1 <- 1 / ((1 - p_x) * v0) + 1 / (p_x * v1)
  lambda_w <- n * b1^2 / Ib1

  # Score
  lambda_s <- n * p_x * (1 - p_x) * (p1 - p0)^2 /
    (p_x * v0 + (1 - p_x) * v1)

  # LRT: equivalent KL-divergence expression for D_star
  p_star <- (1 - p_x) * p0 + p_x * p1
  KL <- function(p, q) {
    p * log(p / q) + (1 - p) * log((1 - p) / (1 - q))
  }
  D_star <- 2 * ((1 - p_x) * KL(p0, p_star) +
                   p_x * KL(p1, p_star))
  M_t <- ((1 - p_x) * v0 + p_x * v1) /
    (p_star * (1 - p_star))
  lambda_lrt <- max(0, n * D_star + 1 - M_t)
```

```
  c(Wald = lambda_w,
           Score = lambda_s,
           LRT = lambda_lrt)
}

## (a) Power at the observed sample size and fitted effect
power_current <- pchisq(critical, df = 1,
                        ncp = ncp_all(n, b1),
                        lower.tail = FALSE)

## 80% and 90% power
lambda_needed <- function(target) {
  uniroot(function(lambda) {
    pchisq(critical, df = 1, ncp = lambda,
           lower.tail = FALSE) - target
  }, interval = c(0, 100), tol = 1e-10)$root
}

lambda_80 <- lambda_needed(0.80)
lambda_90 <- lambda_needed(0.90)

## (b) sample size: hold b0, b1 and p_x fixed
N_needed <- function(test, lambda_target) {
  ceiling(uniroot(function(n_new) {
    ncp_all(n_new, b1)[test] - lambda_target
  }, interval = c(1, 1e6), tol = 1e-8)$root)
}

## (c) MSE OR: hold n, b0 and p_x
direction <- if (b1 >= 0) 1 else -1

MDE_OR <- function(test, lambda_target) {
  effect <- uniroot(function(effect) {
    ncp_all(n, direction * effect)[test] - lambda_target},
    interval = c(0, 2),
    tol = 1e-10)$root
  exp(direction * effect)
}

# Parameters assumed
print(data.frame(n = n,
                 p_x = p_x,
```

```
                  b0 = b0,
                  b1 = b1,
                  p_control = plogis(b0),
                  p_treatment = plogis(b0 + b1),
                  OR = exp(b1),
                  alpha = alpha),
      row.names = FALSE)
```

```
#>     n       p_x          b0        b1 p_control p_treatment     OR alpha
#>  1152 0.9557292 -0.03922071 0.2967657 0.4901961   0.5640327 1.3455  0.05
```

```
# Summary table
results <- data.frame(Test = tests,
                      Power_percent = round(100 * power_current, 2),
                      N_80 = sapply(tests, N_needed,
                                    lambda_target = lambda_80),
                      N_90 = sapply(tests, N_needed,
                                    lambda_target = lambda_90),
                      MDE_OR_80 = round(sapply(tests,
                                               MDE_OR,
                                               lambda_target = lambda_80),
                                        3),
                      MDE_OR_90 = round(sapply(tests, MDE_OR,
                                               lambda_target = lambda_90),
                                        3))

print(results, row.names = FALSE)
```

```
#>   Test Power_percent N_80  N_90 MDE_OR_80 MDE_OR_90
#>   Wald         17.90 8435 11292     2.238     2.542
#>  Score         17.80 8498 11376     2.317     2.692
#>    LRT         17.89 8446 11307     2.253     2.572
```

6 Question 5

Let $Y_j \sim \text{Binomial}(n_j, p_j)$ independently for groups $j = 1, 2$. Under the log-link model,

$$p_1 = \exp(\beta_0), \qquad p_2 = \exp(\beta_0 + \beta^*).$$

We test $H_0 : \beta^* = 0$ against $H_A : \beta^* \neq 0$. This is equivalent to testing $p_1 = p_2$. Define

$$\hat{p}_1 = \frac{Y_1}{n_1}, \qquad \hat{p}_2 = \frac{Y_2}{n_2}, \qquad \tilde{p} = \frac{Y_1 + Y_2}{n_1 + n_2}.$$

The unrestricted maximum likelihood estimate is $\hat{\beta}^* = \log(\hat{p}_2/\hat{p}_1)$. The Fisher information matrix and its inverse are

$$I(\beta_0, \beta^*) = \begin{pmatrix} \dfrac{n_1 p_1}{1-p_1} + \dfrac{n_2 p_2}{1-p_2} & \dfrac{n_2 p_2}{1-p_2} \\ \dfrac{n_2 p_2}{1-p_2} & \dfrac{n_2 p_2}{1-p_2} \end{pmatrix}$$

and

$$I(\beta_0, \beta^*)^{-1} = \begin{pmatrix} \dfrac{1-p_1}{n_1 p_1} & -\dfrac{1-p_1}{n_1 p_1} \\ -\dfrac{1-p_1}{n_1 p_1} & \dfrac{1-p_1}{n_1 p_1} + \dfrac{1-p_2}{n_2 p_2} \end{pmatrix}.$$

6.0.1 a. Wald test

The lower-right entry of I^{-1} gives the estimated variance of $\hat{\beta}^*$. Therefore, the Wald statistic is

$$W = \frac{[\log(\hat{p}_2/\hat{p}_1)]^2}{\dfrac{1-\hat{p}_1}{n_1 \hat{p}_1} + \dfrac{1-\hat{p}_2}{n_2 \hat{p}_2}}.$$

6.0.2 b. Score test

Under H_0, both groups have the common probability $\tilde{p}$. After accounting for the nuisance parameter β_0, the efficient score and its information are

$$U_{\text{eff}} = \frac{Y_2 - n_2\tilde{p}}{1-\tilde{p}}, \qquad I_{\text{eff}} = \frac{n_1 n_2 \tilde{p}}{(n_1+n_2)(1-\tilde{p})}.$$

Hence the score statistic simplifies to

$$S = \frac{U_{\text{eff}}^2}{I_{\text{eff}}} = \frac{(\hat{p}_2 - \hat{p}_1)^2}{\tilde{p}(1-\tilde{p})(1/n_1 + 1/n_2)}.$$

6.0.3 c. Likelihood ratio test

The unrestricted model uses $\hat{p}_1$ and $\hat{p}_2$; the model under H_0 uses $\tilde{p}$ for both groups. Twice the difference between their maximized log likelihoods is

$$G^2 = 2\sum_{j=1}^{2}\left[Y_j \log\left(\frac{\hat{p}_j}{\tilde{p}}\right) + (n_j - Y_j)\log\left(\frac{1-\hat{p}_j}{1-\tilde{p}}\right)\right]$$

Under H_0, each of W, S, and G^2 has an asymptotic χ_1^2 distribution. At significance level 0.05, reject H_0 when the chosen statistic exceeds 3.841. The formulas above assume interior estimated probabilities ($0 < \hat{p}_1, \hat{p}_2, \tilde{p} < 1$).